

率尺度的局部灵敏度均约为  $-0.89$ 。

解析基线、网格收敛、独立有限元和累计守恒均支持上述数值结果。宽情景计算进一步表明，当长期平衡含水率接近 0.15 时，终点会显著延后，甚至超出半径观测时域。实际应用应优先校准  $D(C, T)$ 、平衡含水率和表面传质关系，再结合安全裕度确定停机时间。

## AI 工具使用声明

本参赛队在竞赛过程中使用了 AI 工具，主要用于语言润色、程序审查、数值验证和图表排版，详细使用情况见支撑材料。

## 参考文献

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## 附录

### 附录 A 输入插值留出检验

图10(a)–(c) 比较留出观测与两种插值重建，(d)–(f) 给出逐时刻残差、RMSE、MAE 和最不利留出点，(g) 再用持续值 RMSE 归一化三种不同量纲的输入。线性插值相对